

CV 5101
Modeling, Uncertainty, and
Data for Engineers
(July – Nov 2026)

Dr. Prakash S Badal

Announcements

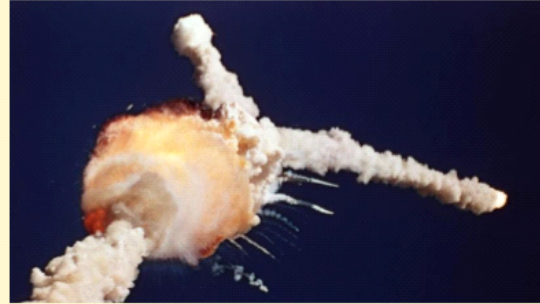
- Part 3 starts from tomorrow (7th Oct)
 - signal processing
 - time series
 - machine learning
- Quiz II on 13th Oct 2026 at 8 AM; venue: STR 301
 - Syllabus: Part-2
 - Slot interchange B / F
 - Slot B class on 13th Oct, if any, will be held at 5 PM

Why should you care?

Why should you care about Risk & Reliability?

- Space Shuttle *Challenger* disaster (1986)

<https://www.youtube.com/watch?v=yibNEcn-4yQ>



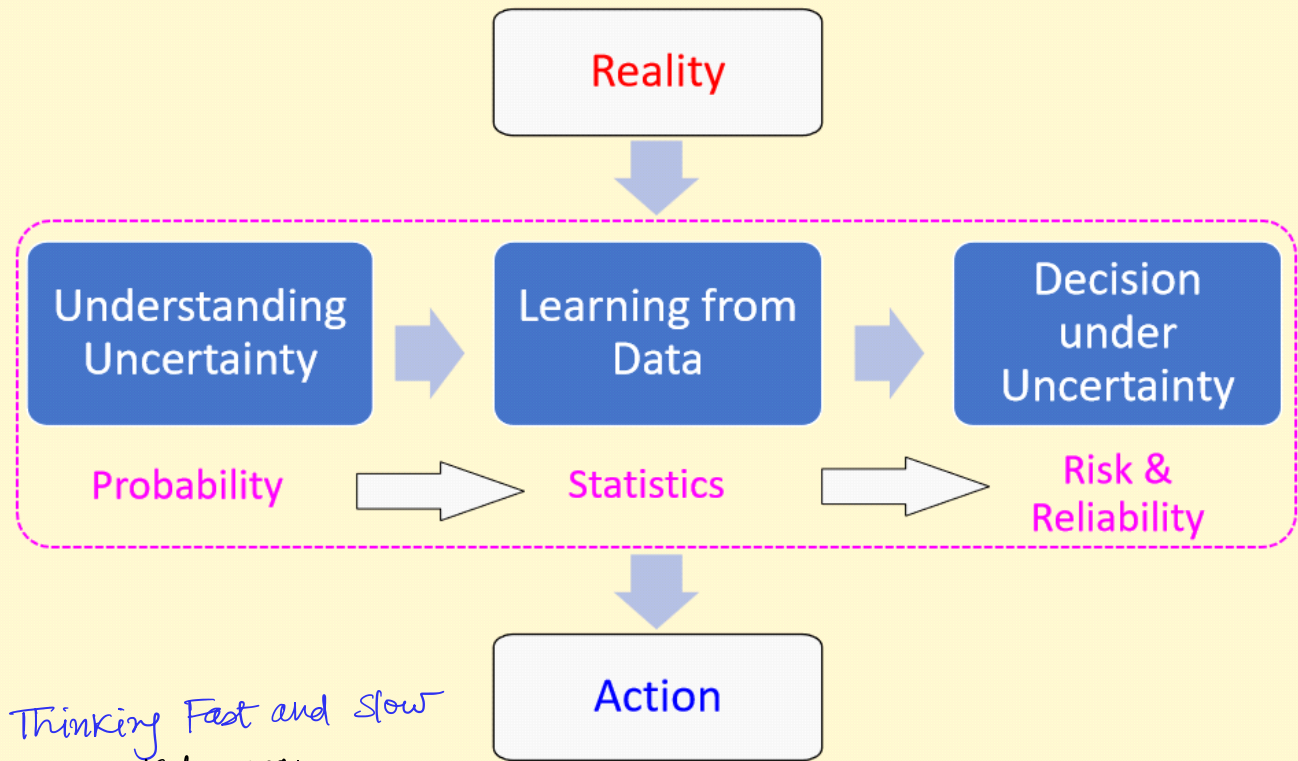
Catch-all approach:

Build for every “what if” scenario: create backup for every imaginable scenario, even wildly unlikely ones

IF EVERYTHING IS IMPORTANT,
THEN NOTHING IS !

} pick your
battles

Three Steps of Every Engineering Problem



Thinking Fast and Slow
Daniel Kahneman

Probability: Understanding Uncertainty

Set Theory and Probability ^{is independence} $\Pr(E_1)$

Conditional probability:

Multiplication rule:

$$\Pr(E_1, E_2) = \Pr(E_1 | E_2) \cdot \Pr(E_2)$$

Statistically Independence:

$$\Pr(E_1, E_2) = \Pr(E_1) \cdot \Pr(E_2)$$

unknown classmate coming to the class

Mutually Exclusive Events:

$$E_i \cap E_j = \emptyset$$

Collectively Exhaustive Events:

$$\bigcup_i E_i = S$$

EQ ← minor
moderate
severe

MECE: → partitioning

Total Probability Rule:

What will happen?

weather @ airport
← clear
light
Heavy
Cyclone

Bayes' Rule:

Now that this has happened,
what caused it?

OJ Simpson's Murder Trial

$\Pr(\text{Guilty} | \text{DNA Match})$

Prosecutor's Fallacy

How come...
what caused it?

$$\Pr(\text{Guilty} | \text{DNA Match})$$
$$\Pr(\text{DNA Match} | \text{Guilty})$$

Prosecutor's Fallacy

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Random Variables

X → childhood variable
 X → Random variable

Definition: Neither Random; Nor a Variable

Given a (random) experiment, a function $\mathcal{S} \rightarrow \mathbb{R}$

Types: Discrete and Continuous → could be countably infinite

PMF, PDF: Mass and Density from Mechanics.

CDF:

Expectation of RV: Expected value from a die roll → Measure of Centrality
Loaded die → weighted average

Variance: Measure of dispersion à la Moment of Inertia à la Center of Mass

Higher moments:

$$\theta = E[(X - \mu)^3] / \sigma^3$$

skewness

$$\kappa = E[(X - \mu)^4] / \sigma^4$$

kurtosis → Tailedness

Covariance: $\text{COV}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$

Correlation: $\rho_{xy} = \frac{\text{COV}(X, Y)}{\sigma_X \sigma_Y}$

$$E[(X - \mu_X)^2]$$

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Distributions

Normal/Gaussian: Ubiquitous; Whenever there are additive effects.

Uniform:

Exponential:

Lognormal:

Gumbel distribution:

$$\min(x_1, x_2, \dots, x_n) \rightarrow$$

$$\max(x_1, x_2, \dots, x_n) \rightarrow$$

a series of links in a chain
Bridge designed for max flood

Correlation:

Correlation:

for max flood

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Uncertainty Quantification

Aleatory: Irreducible, Inherent

Epistemic: Reducible with more data /
better models

Wind speed on building surface example:

" Line between aleatory & epistemic
are somewhat blurry.

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Four Categories of Prediction

Analytical solution:

know the physics and exact solution exists

Approximate solution:

know the physics but exact solution is difficult or does NOT exist
Approximation

Simulation:

know the physics but evaluation of full uncertainty is prohibitive
Simulation, e.g., Monte Carlo

Data-driven prediction:

Learn from the relationship within data.
Reward the prediction

FUTURE: Combine Physics, Simulation, Approximation,
& Data-driven Learning. (INTERFACE)

Mean and Variance propagation laws

- If $Y = g(X)$ is a nonlinear function of X . Find $E[Y]$ & $\text{Var}(Y)$.

$$E[Y] = E[g(X)]$$

$$E[Y] \cong g(\mu_X)$$

First-order mean approximation

$$E[Y] \cong g(\mu_X) + \frac{1}{2} \left(\frac{\partial^2 g}{\partial x^2} \right)_{\mu_X} \sigma_X^2$$

Second-order mean approximation

$$\text{Var}[Y] = E[(Y - \mu_Y)^2] \cong \left(\left(\frac{\partial g}{\partial x} \right)_{\mu_X} \right)^2 \sigma_X^2$$

First-order var. approx.

Regression & estimation

Elements of models

0. Eyeball the data.

Scatter plot, histogram, change scales

1. Estimation **Goal:** Obtain parameter estimates ($\hat{\beta}$)

Concepts: least squares, maximum likelihood, fitting the model

2. Inference **Goal:** Model comparison; uncertainty in parameter ($\hat{\beta}$)

Concepts: Conf. interval for $\hat{\beta}$, hyp. testing, std. error, p-values

3. Prediction **Goal:** Forecast new outcomes (x is now a predictor)

Concepts: CI for $\hat{y}$ (prediction error), mean-squared error (MSE)

4. Explanation **Goal:** Interpret the fitted model, understand relationships

Concepts: feature importance, causality

5. Diagnosis **Goal:** Assess model assumptions and validity

Concepts: error (constant Var.), unusual observations (outliers)

Linear models: example

- Is this a linear model

$$E[Y_i] = \beta_0 \cdot x_{1,i}^{\beta_1} \cdot x_{2,i}^{\beta_2}$$

Write its linear functional relationship

$$E \left(\begin{bmatrix} \ln Y_1 \\ \ln Y_2 \\ \vdots \\ \ln Y_n \end{bmatrix} \right) = \underbrace{\begin{bmatrix} 1 & \ln x_{1,1} & \ln x_{2,1} \\ 1 & \ln x_{1,2} & \ln x_{2,2} \\ \vdots & \vdots & \vdots \\ 1 & \ln x_{1,n} & \ln x_{2,n} \end{bmatrix}}_A \underbrace{\begin{bmatrix} \ln \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}}_x$$

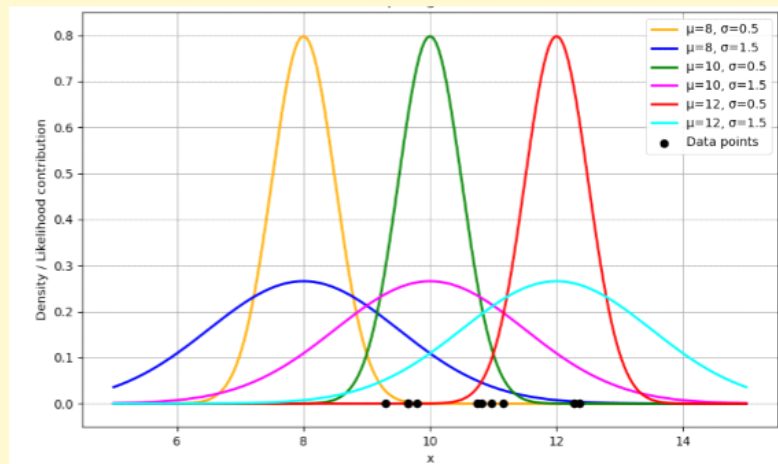
A linear model in x

x_i are coefficients (yet to be determined) in this model

Maximum Likelihood Estimation

Maximum likelihood estimation

$$\hat{\mu}, \hat{\sigma} = \arg \max_{\mu, \sigma} \ln[\mathcal{L}(\mu, \sigma | \mathbf{x})]$$



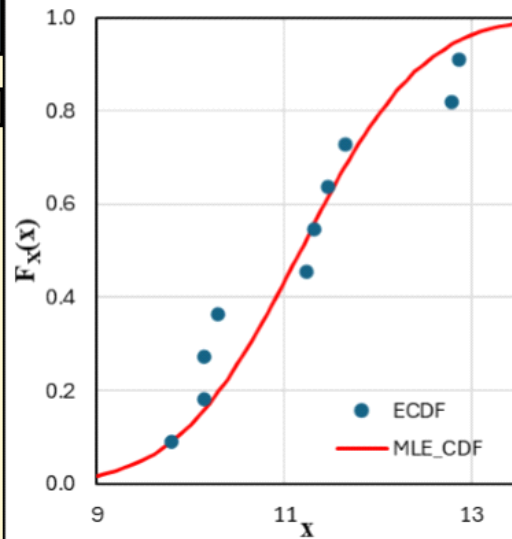
color	oran.	blue	green	mag.	red	turq.
μ_{guess}	8	8	10	10	12	12
σ_{guess}	0.5	1.5	0.5	1.5	0.5	1.5
x_i	Likelihood					
10.7	0.00	0.05	0.26	0.24	0.03	0.19
9.8	0.00	0.13	0.73	0.26	0.00	0.09
11.0	0.00	0.04	0.12	0.22	0.10	0.21
12.3	0.00	0.00	0.00	0.08	0.68	0.26
9.6	0.00	0.15	0.62	0.26	0.00	0.08
9.6	0.00	0.15	0.62	0.26	0.00	0.08
12.4	0.00	0.00	0.00	0.08	0.61	0.26
11.2	0.00	0.03	0.06	0.20	0.19	0.23
9.3	0.03	0.18	0.30	0.24	0.00	0.05
10.8	0.00	0.05	0.21	0.23	0.05	0.19
Sum-log-lik.	-166	-31	-32	-17	-59	-20

$\mu_{\text{goal-seek}}$ **10.67**
 $\sigma_{\text{goal-seek}}$ **1.03**

Lik.	Log-lik.
0.39	-0.4
0.27	-0.6
0.37	-0.4
0.11	-0.9
0.24	-0.6
0.24	-0.6
0.10	-1.0
0.35	-0.5
0.16	-0.8
0.38	-0.4

Use "solver" to **maximize** Sum-log-lik. by changing μ, σ .

Sum-log-lik. **-6.3**

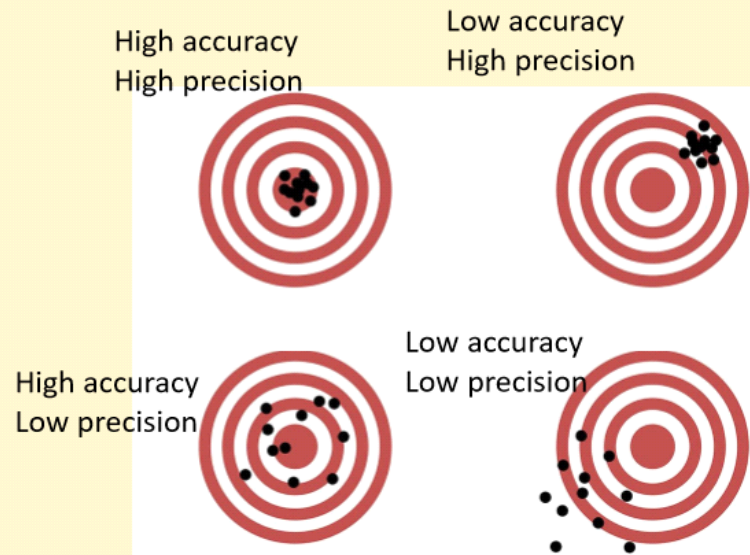


Precision and Bias

Bias and precision

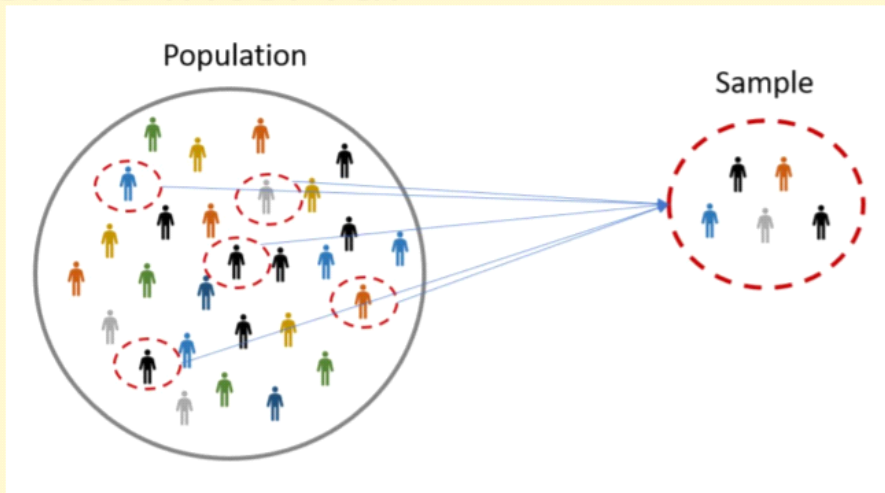
- Accuracy/bias:

- Precision:



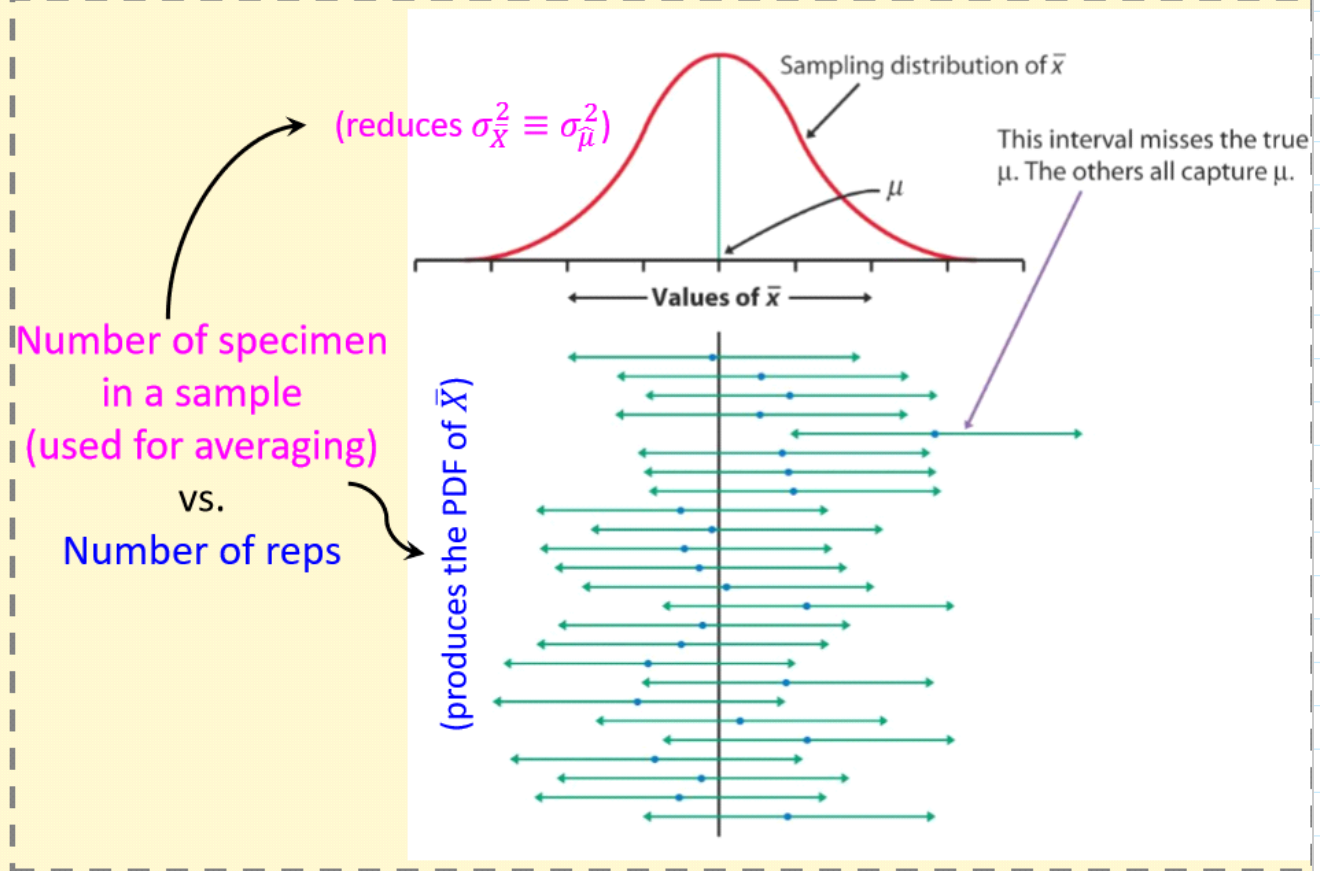
Confidence Interval Objective

Confidence interval



Confidence Interval Estimation

CLT and CI: (estimating mean)

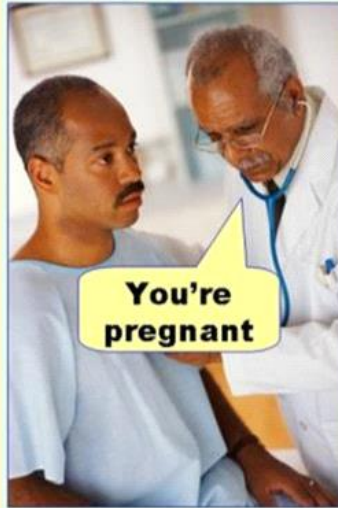


Hypothesis Testing

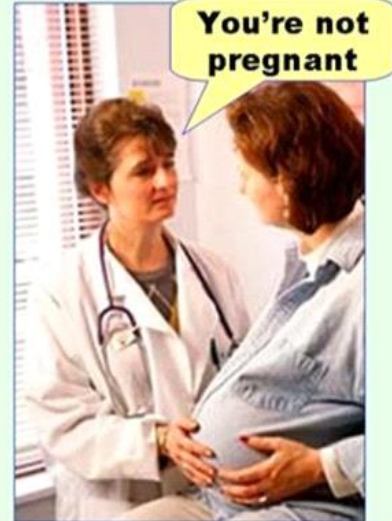
Spot the error*

*: not an example from real life

Type I error
(false positive)



Type II error
(false negative)



Hypothesis testing (& justice system)

No numerical values in courts, but they share four common features:

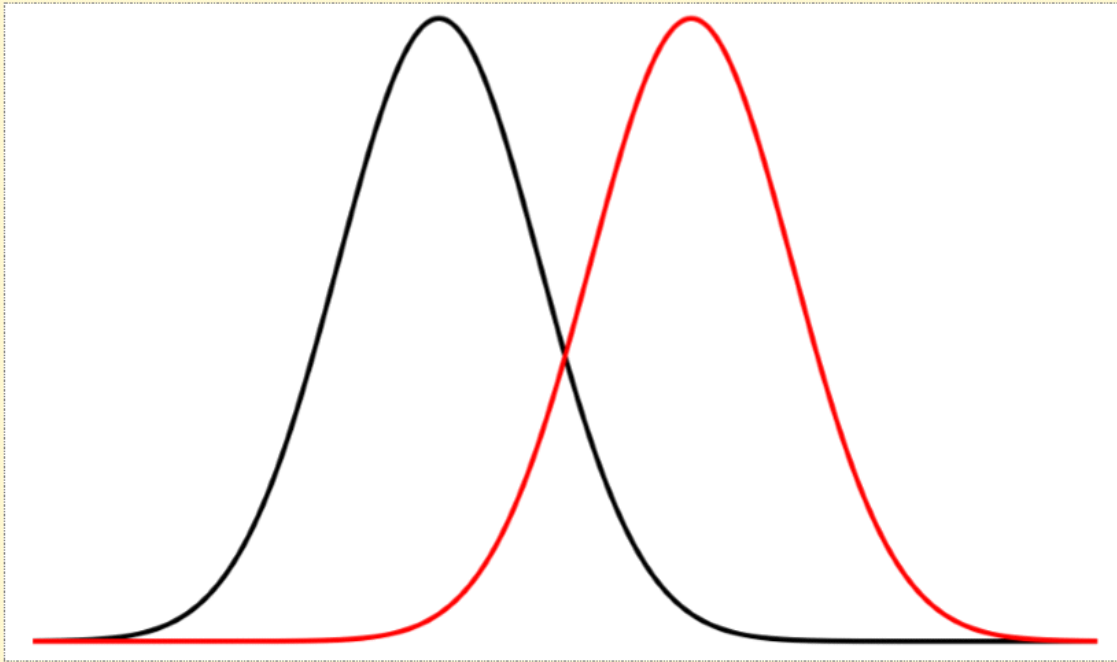
1. The alternative hypothesis:

2. The null hypothesis:

3. A standard of justice:

4. A data sample:

Type II error



Extreme value analysis

- Magnitude of extremes

- Generalized extreme value (GEV) distributions**

- Smallest/largest values?
 - Look out for the tails? Thin/thick? Bounded?
 - Estimate excess Kurtosis
 - Pick one of the GEV models

- Frequency of extremes

- Poisson process**

- Count of extremes $\rightarrow$ Estimate event rate (λ)
 - Independent or clustered?
 - Proceed with Poisson models
 - Decluster using peak-over-threshold/block maxima

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Risk & Reliability

Probabilistic seismic risk assessment

- Three components

Given: location & building design ($\mathcal{L}$ and $\mathcal{D}$)

- Seismic hazard,
- Building's fragility, drift demand,
- Consequence of damage, e.g., downtime, repair cost ratio:

Thank you!

psb@civil.iitm.ac.in